LungMarker: results

Research use only. Not for clinical use. No clinical claims. LungMarker helps a researcher measure lung tumours on CT scans faster. An AI finds the tumours and drafts the measurements; the researcher checks and corrects each one, and the results go straight into the lab's Excel sheet, their one source of truth. Built for the Harbor Thoracic Oncology Lab (simulated) and tested on a public set of CT scans: 59 scans from 31 patients, each with one tumour outlined by an expert (RIDER Lung CT).

What works

52 of 59 found

The AI found the tumour the expert had marked in 52 of 59 scans (88%), with no hints.

59 scans, 31 patients

24 of 24 matched

Every tumour found on both of a patient's scans was correctly matched to the earlier scan.

24 patient pairs

3.0 mm vs 2.3 mm

The AI's measurement of a tumour's longest width is typically 3.0 mm from the expert's; two expert measurements of the same tumour differ by 2.3 mm. No tendency to over- or under-measure.

52 tumours, 28 patients (expert: 28 patients)

What fails, or isn't proven yet

7 misses, all central

All 7 missed tumours were in the centre of the chest, next to the heart and main airways (14–52 mm). The app now requires the researcher to check that area before approving; an automatic warning was tried and dropped (too many false alarms).

7 of 59 scans, 4 of 31 patients

19 of 52 off by > 5 mm

AI outlines more than 5 mm from the expert's (13 too small, 6 too large). The researcher confirms every outline before it's saved, but some large misses get no "AI unsure" warning.

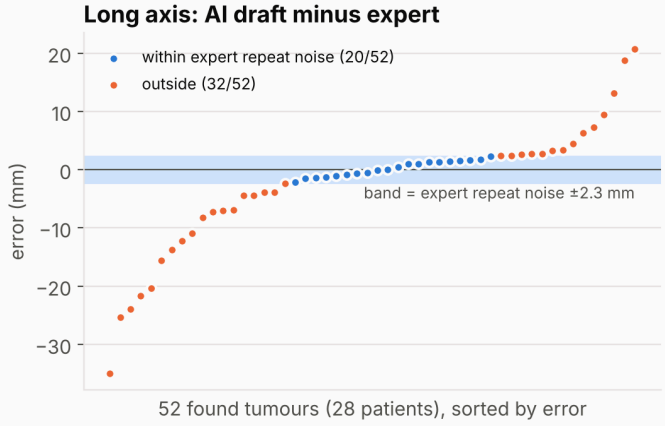
52 tumours, 28 patients

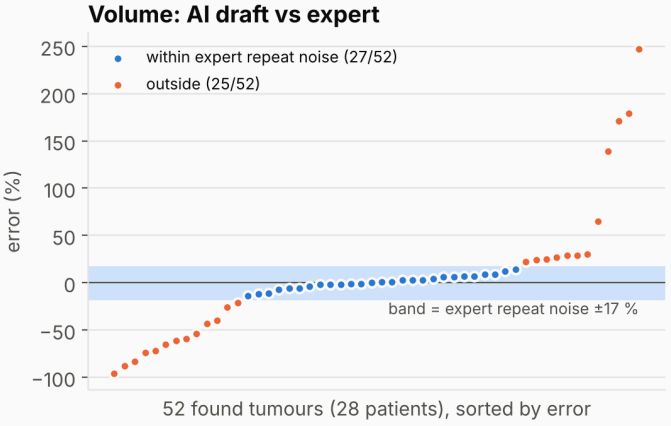
Not tested

Faint "ground-glass" nodules: none in the test scans, and the open set that has them (LNDb) needs approval that takes up to 9 days, ending after the deadline. False alarms: the data marks only one tumour per scan. The lab's own scans.

0 ground-glass nodules in the 59 test scans

AI first draft vs expert outline (RIDER, research use). Every draft is checked by the reviewer before it is recorded.





Each dot is one of the 52 tumours found (28 patients): how far the AI's measurement is from the expert's (left: longest width, mm; right: volume, %). Blue band: how much the expert's own two measurements of the same tumour differ (2.3 mm, 17%; 28 patients). Within the band: 20 of 52 for width, 27 of 52 for volume. A good first draft, not a final answer, which is why the researcher confirms every value.

How I measured the time saved

The honest test is the real researcher timing his work with and without the tool. Because the lab's users are simulated, that wasn't possible. Instead I combined what I could collect, agreed the method with Base44's FDE lead (Sam), and labelled every number as an estimate.

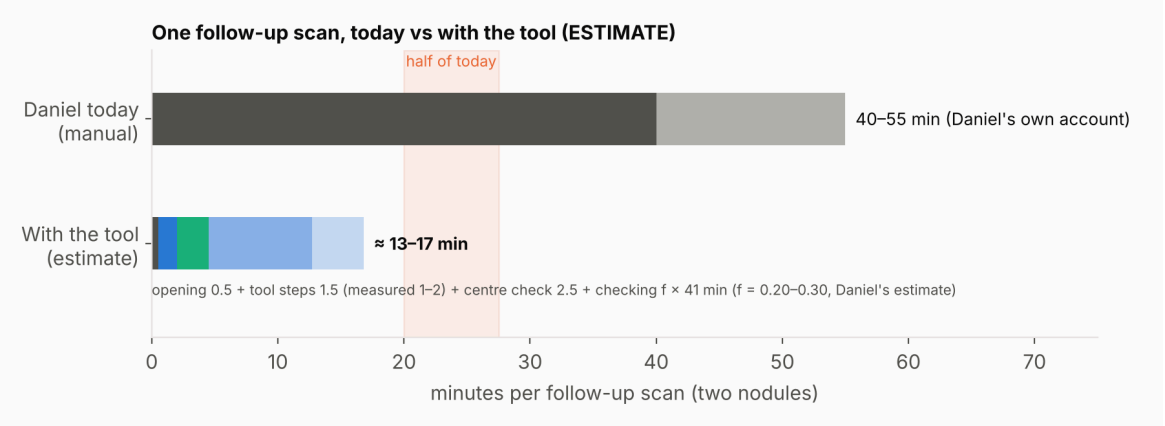
Today's time per incoming scan. Two-thirds of the scans Daniel receives are follow-ups, which take longer than first (baseline) scans, so the average is weighted:

Minutes per scan = ⅓ × baseline + ⅔ × follow-up = ⅓ × 39 + ⅔ × 47.5 ≈ 45 min

Baseline 39 min: Daniel's log (4 scans). Follow-up 40–55 min, midpoint 47.5: Daniel's own step-by-step account (his log: 55 min, 1 scan; using it gives ≈ 50 min, so we use the lower figure). The lab's target, half: about 22 min per scan.

Where the time goes today (one follow-up scan), and what the tool changes

Step	Daniel today (his account)	What the tool does	Effect
Opening the scan	1–2 min	Opens with the AI's suggestions ready; earlier scan already linked	Shortened
Finding each earlier nodule and matching it	~10 min per nodule	Suggests the match and shows both scans side by side; he confirms	Shortened, still checked
Searching for new nodules	10–20 min (two passes, 250–400 images)	AI marks candidates; one check pass, plus the required centre-of-chest check	Search shortened; check kept
Measuring	2–4 min per nodule	Measurements drafted; he checks or corrects them	Shortened, still checked
Full outline	15–25 min per tumour, when needed	Outline drafted for every nodule; he confirms	Shortened
Typing into Excel	3–5 min	One export a day in his format	Removed
New	—	Accept or reject each AI suggestion; tick the centre-of-chest check	Added
Total	**40–55 min**	**With the tool: ≈ 13–17 min = opening 0.5 + checking (f × 41) + tool steps 1.5 (measured 1–2) + centre check 2.5; f = 0.20–0.30 (his estimate)**	



With the tool (estimate). Two nodules: opening 0.5 + checking (f × 41) + the tool's own steps 1.5 (measured 1–2 min, 7 runs; Annex C) + central check 2.5. 41 min = searching 15 (midpoint of 10–20) + matching 2 × 10 + measuring 2 × 3. f is Daniel's estimate that checking takes "about 20–30% of the time" of doing it from scratch, "assuming the draft is generally solid" (Slack, 20 Oct). Result: **≈ 13–17 min vs 40–55 today, roughly 60–75% less**; still half as long as f < **~0.45**. Not a measurement, and not every draft is solid: 19 of 52 outlines were more than 5 mm off and central tumours are missed, so those cases take longer.

Limitation and next step. The lab's users are simulated: they answer questions but can't open the tool, so no real reader was timed (checking time is Daniel's estimate) and no real usability test was possible. Daniel approved the flow from screenshots and one external tester (a doctor) tried the tool. The measurement itself is built in: the app times every scan the way Daniel logs it today (the whole review, without long breaks) and fills his "Minutes on scan" column. **Next step:** Daniel's own timed reviews, with and without the tool, on a small set of the lab's scans, to replace his estimate with a measurement.

Annex A · Method

- **Data:** RIDER Lung CT (public, The Cancer Imaging Archive): 31 patients, each scanned twice about 15 minutes apart; 59 scans with an expert outline of one tumour each (RIDER-LungCT-Seg). Tumours 13–93 mm (median 43 mm), mostly solid (53 solid, 6 part-solid), no ground-glass. The lab measures nodules from 4 mm, so these are larger and clearer than its usual cases.
- **How the AI ran:** the whole pipeline with no hint: lobes, then nodule finding (TotalSegmentator `lung_nodules`, full settings, free Colab T4 GPU, median 49 s per scan), then outline and measurement, then matching between each patient's two scans. The finding model was not trained on RIDER.
- **How quality was measured:** the AI's outline and the expert's outline of the same tumour are measured the same way, the way Daniel measures: longest width on the slice where the tumour is largest, the longest perpendicular width on that slice, and volume from the 3D outline. Error = AI minus expert, per tumour; we report the median of the absolute error and the median of the signed error (bias).
- **Expert noise (the yardstick):** the expert outlined the same tumour on both of a patient's scans, taken 15 minutes apart. The median difference between those two outlines (28 patients) is 2.3 mm for longest width and 17% for volume. The AI is "within expert noise" when its error is no larger.
- **Outline setting:** the model's core plus a rim scaled to the core's size (the shipped setting). It was chosen by comparing the model's own outline, MedSAM 2 and four rim variants on these same 52 tumours, so the quality numbers are somewhat optimistic.
- **What this measures:** the AI's first draft. The researcher's confirmed values (what goes into the sheet) were not measured.

Annex B · Full numbers

Measure	Sample	Result
Expert tumour found	59 scans, 31 patients	52 / 59 (88%); solid 48 / 53, part-solid 4 / 6
Missed tumours	59 scans	7 scans, 4 patients; all central, 14–52 mm
Longest width, AI vs expert	52 tumours, 28 patients	median error 3.0 mm, bias −0.3 mm; 20 / 52 within 2.3 mm; 19 / 52 more than 5 mm off (13 too small, 6 too large)
Volume, AI vs expert	52 tumours, 28 patients	median error 14%, bias +0.5%; 27 / 52 within 17%
Overlap with expert outline (Dice)	52 tumours	median 0.76
Matched to the earlier scan	24 patient pairs	24 / 24; alignment error at the tumour median 2.9 mm (max 13.9)
Lobe agrees with expert outline	52 tumours	52 / 52
Other AI marks per scan	59 scans	median 3; 82% reappear on the patient's other scan (a proxy for false alarms only)
Small nodules (< 8 mm), same nodule on both scans	84 pairs, 22 patients	width differs by median 0.5 mm; volume by ~30%
"AI unsure about the edges" on likely-too-small outlines	52 tumours; 322 AI marks	flags all 4 small faint tumours drawn much too small; fires on 12% of all marks (tuned on 4 cases)
Automatic "check the centre" warning (dropped)	59 scans	best version flagged 4 of the 7 missed scans, but also 49 of the other 52
Tool's own steps (developer pilot)	7 scans, 4 patients	1–2 active min per scan (median 2)
Follow-up with the tool (estimate)	Daniel's estimate f = 0.20–0.30	0.5 + f × 41 + 1.5 + 2.5 ≈ 13–17 min (two nodules); 60–75% less; break-even f ≈ 0.45 (0.38–0.56)

Annex C · All 8 developer runs

Patient (public RIDER)	Timepoint	Active min	Open min	Nodules reviewed	Note
RIDER-1225316081	Baseline	1	1	2	
RIDER-1225316081	Follow-up 1	*12*	*20*	3	**Excluded:** scan left open while writing messages, so the timer measured something else
RIDER-1332496276	Baseline	1	1	2	
RIDER-1332496276	Follow-up 1	1	1	2	
RIDER-1500037140	Baseline	2	2	4	
RIDER-1500037140	Follow-up 1	2	2	4	
RIDER-1532432635	Baseline	2	2	5	incl. 2 rejected
RIDER-1532432635	Follow-up 1	2	2	5	incl. 1 rejected + 1 not found
Included: 7 scans		**1–2 (median 2)**		**2–5**	**developer, not a trained reader**

18 Oct, final app version, by the developer (Stav). Source: the app's own timer ("Minutes on scan" and the approval rows of the edit trail in the Excel export). Active minutes pause after 2 minutes without input or when the tab is hidden; whole minutes, so "1" means roughly 0.5–1.5 min. These runs measure the tool's mechanics, not a trained reader's judgement.

Annex D · Daniel's step breakdown (his words, Slack, 18 Oct)

Step (follow-up scan)	Minutes	Unit
Loading	1–2	per scan
Re-finding and matching (aligns by bronchi and fissures; slice numbers are unreliable because of breathing)	~10	per nodule
Finding new nodules (two passes over 250–400 slices)	10–20	per scan
Measuring	2–4	per nodule
Full outline	15–25	per tumour, when needed
Recording	3–5	per scan
Total (his own figure)	**40–55**	**per scan**

Some steps are per nodule, so the total is Daniel's own figure, not a sum we calculated. His log ("Minutes on scan", last 5 rows): baseline 38, 49, 30, 37 (mean ≈ 39); follow-up 55. About two-thirds of incoming scans are follow-ups (his estimate). He has not broken down a baseline scan.

Annex E · Credits

- **RIDER Lung CT:** Zhao, B. et al. (2015). *Coffee-break lung CT collection with scan images reconstructed at multiple imaging parameters.* The Cancer Imaging Archive. doi:10.7937/k9/tcia.2015.u1×8a5nr (CC BY 3.0).
- **RIDER-LungCT-Seg:** Wee, L., Aerts, H., Kalendralis, P., & Dekker, A. (2020). *RIDER Lung CT Segmentation Labels.* The Cancer Imaging Archive. doi:10.7937/tcia.2020.jit9grk8.
- **TotalSegmentator** (Wasserthal, J. et al., 2023), including the `lung_nodules` model by BlueMind AI. Apache-2.0.
- **MedSAM 2** (Ma, J. et al., 2025, arXiv:2504.03600), compared, not used. **SimpleITK** for lung alignment.
- **Base44** for the review app. Full code, raw data and the long version of these results (`RESULTS.md`) are in the repository; `python results/make_figures.py` rebuilds every figure and number.